EXAMINATION

Luleå University of Technology

Course: ***D7024E***
Course name: ***Distributed and Mobile Computing Systems***
Date: ***2018-10-26 8:00 to 2018-10-30 12:00 (home exam)***
Time: ***Expected time spent is 3-4 hours***
Aid: ***WWW (no interactive communication with others)***

Teacher on duty (telephone number):	***Teacher on duty (complete telephone number):***
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Grade scale (preliminary): ***50% grade 3, 75% grade 4 and 90% grade 5***	Note that this grade is for the exam. The grade for the course may be different as it is weighted on several parts.
Total number of questions and score: ***3 questions, total score 36***	
Other information: This is an individual home exam.	Note: No interactive cooperation is allowed. Information can be searched on the world wide web or in published books/papers. Please provide references, when such information is used in your answers.

General instructions

You shall formulate answers in your own words, crisp and to the point. Avoid straight copy-paste of text from the web (figures may be ok to copy if they illustrate what you describe). Always, please provide references to your sources (e.g., URLs).

Your responses must be uploaded in Canvas before the deadline.

Since there are several other parts in the course, this exam is not intended to cover the complete course content, just to take some samples.

After examination

Examination results will be posted in Canvas within 20 workdays after the examination.

1. Mixed question on distributed systems (11p)

These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must provide a short explanation to get credits. For “true” or “false”, no explanation is required, unless you want to explain what conditions are assumed. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p.

a. Communication in geographically distributed systems is typically synchronous

b. Peer to peer solutions are typically cost effective with respect to needed service capacity.

c. Marshaling (a k a serialization) is not needed in client-cloud solutions, which has made them more popular than peer-to-peer.

d. NAT/firewall traversal by STUN is initiated by external nodes to find addresses of nodes that are behind that firewall.

e. The purpose of TURN is that firewalls can turn incoming DoS attacks back towards the original senders.

f. In typical relational databases there is a transaction log (or command log) to locally handle potential loss of RAM cache state.

g. A replicated (global) command log ensures that commands are ACID.

h. In global consensus solutions, a typical key requirement is that new nodes can automatically join in order to scale out and improve write performance.

i. In the general case, to obtain global consensus you must make assumptions regarding connectivity, or fraction of nodes being alive.

j. Non-Byzantine global consensus solutions typically assume that there can be a small number of traitors.

k. The pub-sub paradigm is typically based on channels or streams of immutable messages.

l. Kafka uses a global consensus solution to keep track of responsibilities for stream spread and replication over the participating Kafka servers

m. Master-slave architectures are often used for scalability where fast and immediate consistency is required.

n. Block chains are suitable for implementing decentralized ledgers.

o. Block chains can represent a distributed state machine.

p. Block-chains supporting generic smart contracts can typically execute many contracts in parallel.

q. Data storage based on consistent hashing place data so that short round trip times are obtained.

r. Distributed hash tables are decentralized and scale better than logical-ring based consistent hashing.

s. Map-reduce aims for that code execution (trough defined procedures) is carried out where the data is stored, in order to reduce communication overhead.

t. Map-reduce solutions like Hadoop are typically geo-distributed.

u. Security by PGP uses public/private key pair cryptography for encrypting and exchanging symmetric keys between parties.

v. Network level security (e.g., IPSEC) can be used to create encrypted tunnels

Note: In the following questions 2 and 3, it is important to include sources (e.g., URLs) and that you explain in your own words.

2. Distributed systems properties (12p)

a) In distributed computing systems, which properties do we refer to in the term horizontal scalability? (4p)
b) In distributed computing systems, how is high availability provided (a conceptual explanation is enough)? Provide an example. (4p)
c) Explain trade-offs between cluster-based systems and geo-distributed systems? (4p)

3. Blockchains (13p)

a) In Bitcoin, explain schematically the proof-of-work procedures, and explain the purpose of this in the consensus process. (5p)
b) In Ethereum, how are multiple smart contracts executed and verified by many nodes, in order to obtain consensus on the correctness of each new block and consensus on the final state? (8p)

Finally, please fill out the course evaluation here:
https://evasys.ltu.se/evasys/online.php?p=QKN4H